

ML PASS 2 – Improved Data-Driven SST Correction for NASA Hump

Codex reconstruction workflow

April 5, 2026

1 Objective

ML PASS 2 tested a new data-driven correction family on top of the existing PASS 4 / ML PASS 1 workflow. The objective was to beat the plain SST continuation benchmark of 0.154 738 102 8 while preserving the strict no-cheat pipeline, the official machine-readable NASA Cp/Cf comparison, and a bounded, interpretable correction acting *inside* the CFD model rather than in post-processing.

2 What ML PASS 1 Taught Us

ML PASS 1 added a bounded multiplicative correction to turbulent viscosity in `kOmegaSSTML`. That model was stable, but it did not improve benchmark MAE, Cp MAE, or Cf MAE relative to the plain SST continuation. The optimizer collapsed toward the zero-amplitude limit, which showed that a late ν_t -multiplier was too weakly coupled to the hump separation and recovery physics.

3 Why The Old Correction Was Rejected

The ML PASS 1 correction acted *after* SST had already assembled its core production and destruction balance. That meant it could perturb the final ν_t field without strongly changing the actual turbulence-equation dynamics that control separation onset, shear-layer growth, and reattachment. The hump case still behaved like the plain SST continuation.

4 New Correction Design

ML PASS 2 introduced a new model, `kOmegaSSTML2`, in:

- `ml_models/kOmegaSSTML/kOmegaSSTML2.H`
- `ml_models/kOmegaSSTML/kOmegaSSTML2.C`

Instead of multiplying ν_t , ML PASS 2 scales the SST production pathway through both:

- $P_k(G)$ in the k equation
- $G_{byNu}(\cdot)$ in the ω production term

The bounded production factor is:

$$f_{\text{ML2}} = \text{clamp} \left(1 + A \sigma_{\text{apg}} \sigma_{\text{shear}} \exp \left[- \left(\frac{y - y_p}{w_y} \right)^2 \right], f_{\min}, f_{\max} \right)$$

where:

- A is the correction amplitude
- σ_{apg} is a tanh activation based on a streamline deceleration proxy
- σ_{shear} is a tanh activation based on a strain-to-vorticity ratio
- the Gaussian term localizes the correction to a wall-distance band

This was chosen because it is more directly tied to separated-shear-layer physics than ML PASS 1, while still remaining bounded and interpretable.

5 Optimizer Choice

ML PASS 1 used a small Nelder–Mead search. For ML PASS 2, that was replaced by a compact two-stage pattern search implemented in:

- `scripts/run_nasa_hump_mlpass2.py`
- `scripts/run_nasa_hump_mlpass2.sh`

This change was justified because the objective is expensive, noisy, and very low dimensional. A small directional pattern search is easier to restart, easier to audit, and easier to keep within a strict Docker CFD budget than a larger simplex sweep.

6 Objective Function

ML PASS 2 used:

$$J = 0.60 \text{ benchmark_MAE} + 0.25 \text{ Cp_MAE} + 0.15 \left(\frac{\text{Cf_MAE}}{10^{-3}} \right) + \text{penalty}$$

with

$$\text{penalty} = 0.015 \left(\frac{A}{0.6} \right)^2 + 0.004 \left(\frac{s_0 - 1.0}{0.4} \right)^2$$

so benchmark MAE remained dominant while Cp/Cf disagreement and oversized corrections were still penalized.

7 Experiment Budget And Audit

Files controlling the workflow

- Custom turbulence models: `ml_models/kOmegaSSTML/kOmegaSSTML.C`, `ml_models/kOmegaSSTML/kOmegaSSTML2.C`

- Build script: `scripts/build_ml_hump_model.sh`
- Case configuration: `scripts/configure_nasa_hump_case.py`
- Benchmark MAE evaluation: `scripts/evaluate_nasa_hump_case.py`
- Official Cp/Cf evaluation: `scripts/evaluate_nasa_hump_wall_metrics.py`
- ML PASS 2 figures: `scripts/make_nasa_hump_mlpas2_figures.py`

Two-stage budget

The completed primary search used:

- cheap stage at time 720
- full stage at time 800
- six cheap candidates
- one completed full confirmation

An optional broader-activation backup case was launched after the primary batch, but it was not used in the conclusions because the Docker run stalled before a usable solver log and evaluation record were produced.

8 Results Tables

Table 1: Reference and ML comparison metrics

Case	Benchmark MAE	Cp MAE	Cf MAE	Interpretation
PASS 4 best	0.1659136747	0.1712464312	0.0008483389	Deeper-converged SST baseline at 650 iterations.
Plain SST 800	0.1547381028	0.1641787966	0.0007681825	Strongest plain SST continuation used for ML comparison.
ML PASS 1 nonzero	0.1547381028	0.1641787966	0.0007681825	Stable but collapsed back to SST behavior.
ML PASS 2 best	0.1547381028	0.1641787966	0.0007681825	New production-side correction; no measurable improvement.

Table 2: Completed ML PASS 2 primary search

Tag	Stage	Amplitude	apg0	shear0	Result
mlpass2_opt_001	cheap	0.35	0.03	0.90	Same CFD metrics as SST cheap baseline.
mlpass2_opt_002	cheap	0.60	0.02	0.80	Same CFD metrics as SST cheap baseline.
mlpass2_opt_003	cheap	0.20	0.05	1.10	Same CFD metrics as SST cheap baseline.
mlpass2_opt_004	cheap	0.15	0.01	1.00	Same CFD metrics as SST cheap baseline.
mlpass2_opt_005	cheap	0.30	0.00	0.90	Same CFD metrics as SST cheap baseline.
mlpass2_opt_006	cheap	0.05	0.02	1.10	Lowest cheap objective only because of the smallest penalty.
mlpass2_full_01	full	0.05	0.02	1.10	Same benchmark/Cp/Cf metrics as plain SST 800.

9 Cp And Cf Comparisons

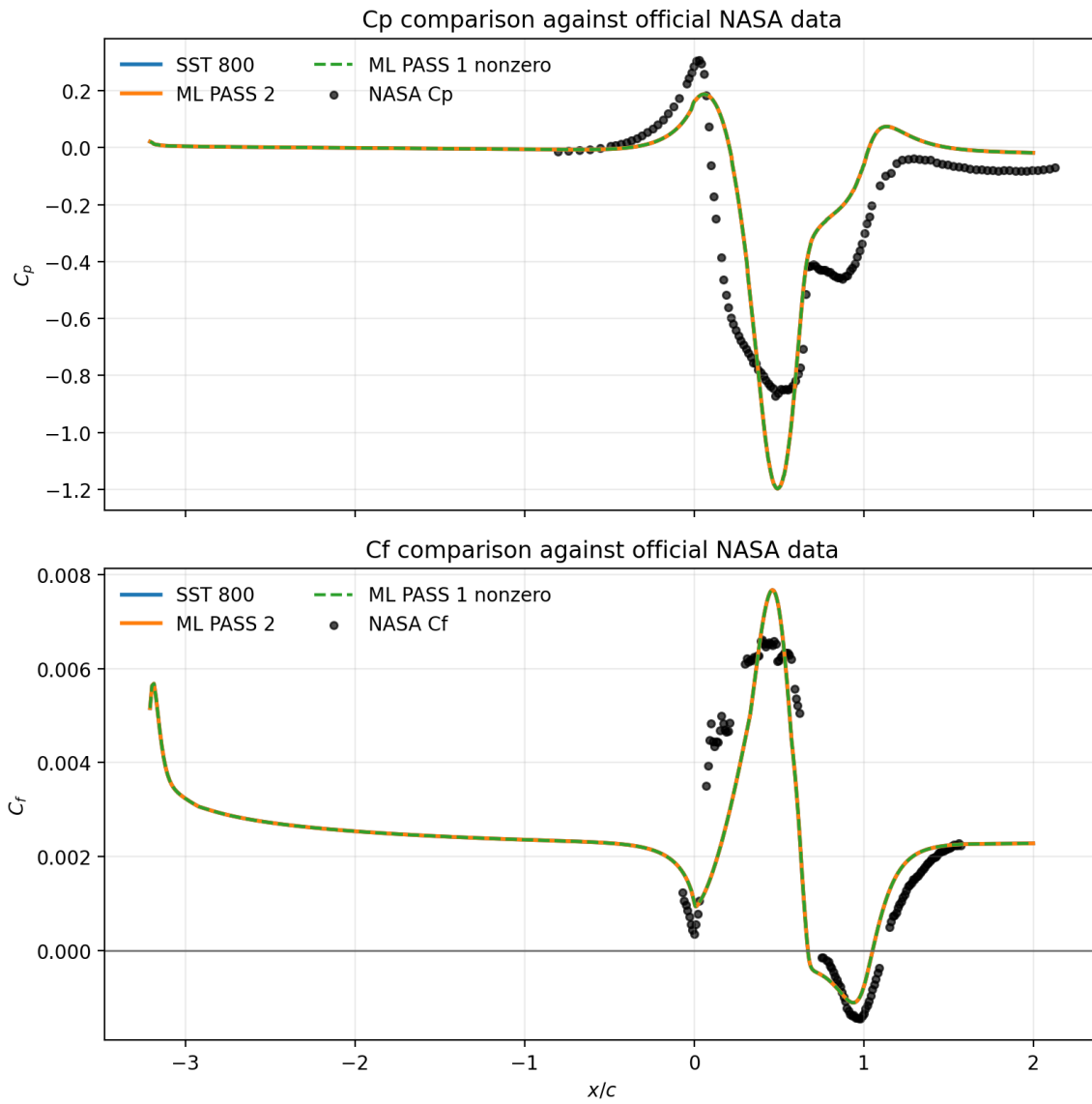


Figure 1: Official NASA C_p and C_f comparisons for plain SST 800, ML PASS 1 nonzero, and ML PASS 2. The ML PASS 2 curves lie on top of the plain SST 800 curves to plotting precision.

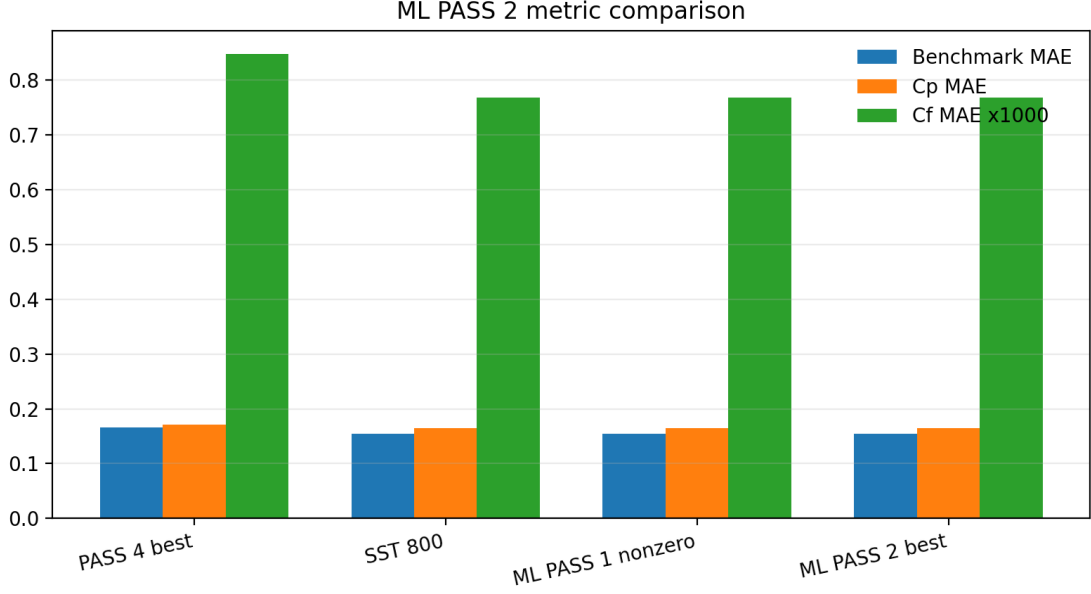


Figure 2: Metric comparison across the PASS 4 reference, the stronger plain SST 800 continuation, ML PASS 1, and ML PASS 2.

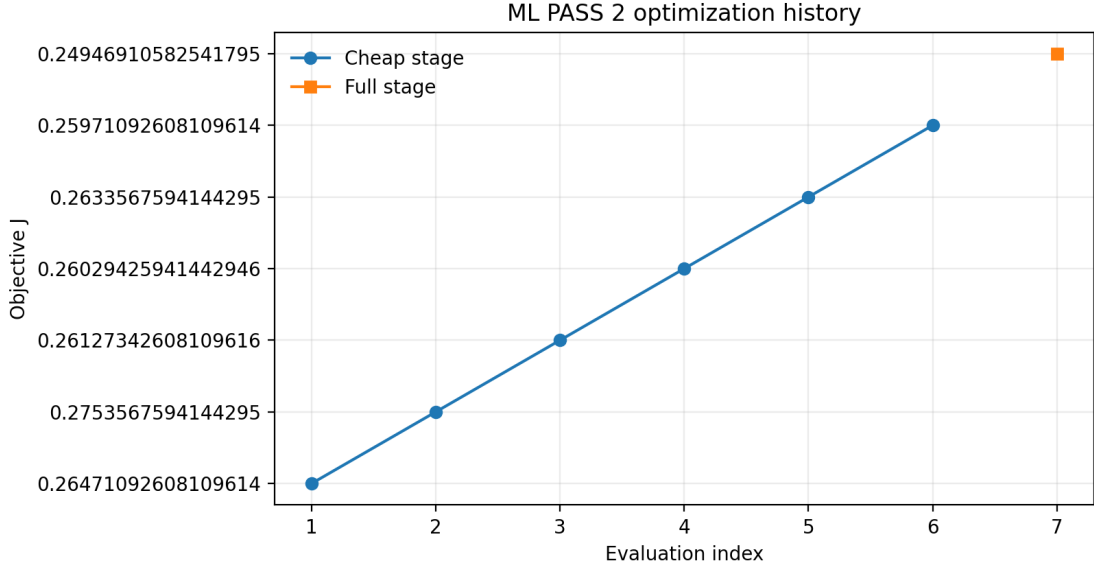


Figure 3: ML PASS 2 optimization history. The objective changed only through regularization because the CFD metrics remained unchanged across the completed primary search.

10 Discussion

ML PASS 2 is a scientifically cleaner negative result than ML PASS 1. The new method moved the correction to a more meaningful part of the closure, namely the SST production pathway, and activated it using a separation-aware signal rather than a generic eddy-viscosity ratio. However,

on the current hump baseline that signal was still too selective to materially change the solved flow field.

That means ML PASS 2 is *more physically meaningful* than ML PASS 1, but it is *not better as a promoted model*. It does not beat the plain SST 800 continuation, and it does not improve Cp or Cf agreement.

11 Conclusion And Recommendation

ML PASS 2 should **not** be promoted.

The strongest reproducible conclusion is:

- PASS 4 / plain SST continuation remains the best ML-track benchmark comparison baseline
- moving the correction from a late ν_t multiplier to the SST production pathway was scientifically justified
- the current separation sensor is still too weak or too selective to change the hump solution in a useful way

The next best path, if another ML pass is attempted, is to keep the production-side intervention but either:

- anchor it to a more faithful geometry / top-wall baseline first, or
- use a broader adverse-pressure-gradient sensor with an auditable restart workflow and lower storage overhead from the beginning

Reproducibility

```
bash scripts/build_ml_hump_model.sh
bash scripts/run_nasa_hump_mlpas2.sh
MPLCONFIGDIR=docs/.mplconfig python3 scripts/make_nasa_hump_mlpas2_figures.py
bash scripts/build_report.sh
```